Homework 07 - Analysis of Variance

Each of the following problems uses the butterfat data set from the faraway library. This dataset provides data on the butterfat content of milk from Canadian cows of five different breeds. For the sake of this assignment, we will consider only mature cows.

```
using RData, DataFrames, CategoricalArrays, GLM, Statistics, StatsPlots, Distributions
```

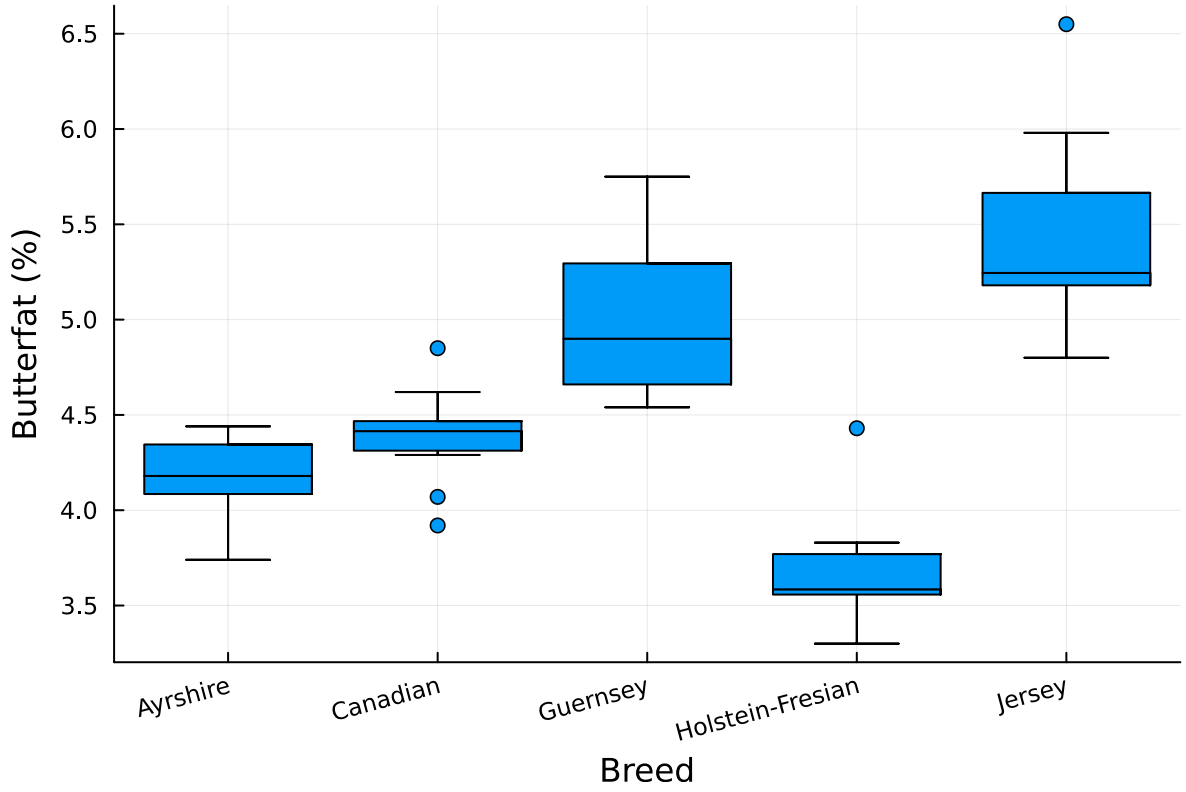
	Butterfat	Breed	Age
1	3.74	"Ayrshire"	"Mature"
2	3.77	"Ayrshire"	"Mature"
3	4.1	"Ayrshire"	"Mature"
4	4.27	"Ayrshire"	"Mature"
5	4.11	"Ayrshire"	"Mature"
6	4.44	"Ayrshire"	"Mature"

```
begin
    # butterfat data set from the faraway R package (CRAN)
    tgz = download("https://cran.r-
    project.org/src/contrib/Archive/faraway/faraway_1.0.8.tar.gz")
    dir = mktempdir()
    run(`tar xzf $tgz -C $dir faraway/data/butterfat.rda`)
    butterfat = load(joinpath(dir, "faraway/data/butterfat.rda"))["butterfat"]
    data1 = subset(butterfat, :Age => a -> a .== "Mature")
    first(data1, 6)
end
```

Problem 1

Part a)

Plot the data by breed and interpret what you see.




```
boxplot(string.(data1.Breed), data1.Butterfat, legend=false, xlabel="Breed",
ylabel="Butterfat (%)", xrotation=15)
```

Jersey and Guernsey have the highest butterfat, Holstein-Fresian the lowest, and Ayrshire and Canadian are similar. Jersey and Guernsey also have more spread.

Part b)

For each of the breeds, find the average butterfat content and find the standard deviation for the butterfat content. Present your solution in a dataframe. Based on the values are there any concerns regarding running regression on the data?

```
stats =
```

	Breed	Mean	SD
1	"Ayrshire"	4.154	0.24627
2	"Canadian"	4.389	0.260532
3	"Guernsey"	5.001	0.408315
4	"Holstein-Fresian"	3.676	0.310133
5	"Jersey"	5.452	0.506377

```
stats = combine(groupby(data1, :Breed), :Butterfat => mean => :Mean, :Butterfat => std
=> :SD)
```

Slight concern: the largest SD (Jersey, 0.51) is about twice the smallest (Ayrshire, 0.25), so equal variance is borderline.

Part c)

Using the effect coding method, create a regression equation for the butterfat content of the different cow breeds. Provide the summary output from R. Based on the summary output show how the average butterfat can be obtained for each of the cow breeds. (Note: Breed 1= Ayrshire, Breed2 = Canadian, Breed3 = Guernsey, Breed4 = Holstein-Fresian) Finally, what does the intercept represent in this model?

```
model =
StatsModels.TableRegressionModel{LinearModel{GLM.LmResp{Vector{Float64}}, GLM.DensePredChol{
```

```
Butterfat ~ 1 + Breed

Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	4.5344	0.0509057	89.07	<1e-51	4.43187	4.63693
Breed: Ayrshire	-0.3804	0.101811	-3.74	0.0005	-0.585459	-0.175341
Breed: Canadian	-0.1454	0.101811	-1.43	0.1602	-0.350459	0.0596585
Breed: Guernsey	0.4666	0.101811	4.58	<1e-04	0.261541	0.671659
Breed: Holstein-Fresian	-0.8584	0.101811	-8.43	<1e-10	-1.06346	-0.653341

```
model = lm(@formula(Butterfat ~ Breed), data1; contrasts = Dict(:Breed =>
EffectsCoding(base = "Jersey")))
```

```
β = [4.5344, -0.3804, -0.1454, 0.4666, -0.8584]
```

```
β = coef(model)
```

	Breed	Computation	Mean
1	"Ayrshire"	"β₀ + β₁"	4.154
2	"Canadian"	"β₀ + β₂"	4.389
3	"Guernsey"	"β₀ + β₃"	5.001
4	"Holstein-Fresian"	"β₀ + β₄"	3.676
5	"Jersey"	"β₀ - β₁ - β₂ - β₃ - β₄"	5.452

```
DataFrame(Breed = levels(data1.Breed),
    Computation = ["β₀ + β₁", "β₀ + β₂", "β₀ + β₃", "β₀ + β₄", "β₀ - β₁ - β₂ - β₃ -
    β₄"],
    Mean = [β[1] + β[2], β[1] + β[3], β[1] + β[4], β[1] + β[5], β[1] - sum(β[2:5])])
```

Butterfat = 4.5344 − 0.3804·Breed1 − 0.1454·Breed2 + 0.4666·Breed3 − 0.8584·Breed4

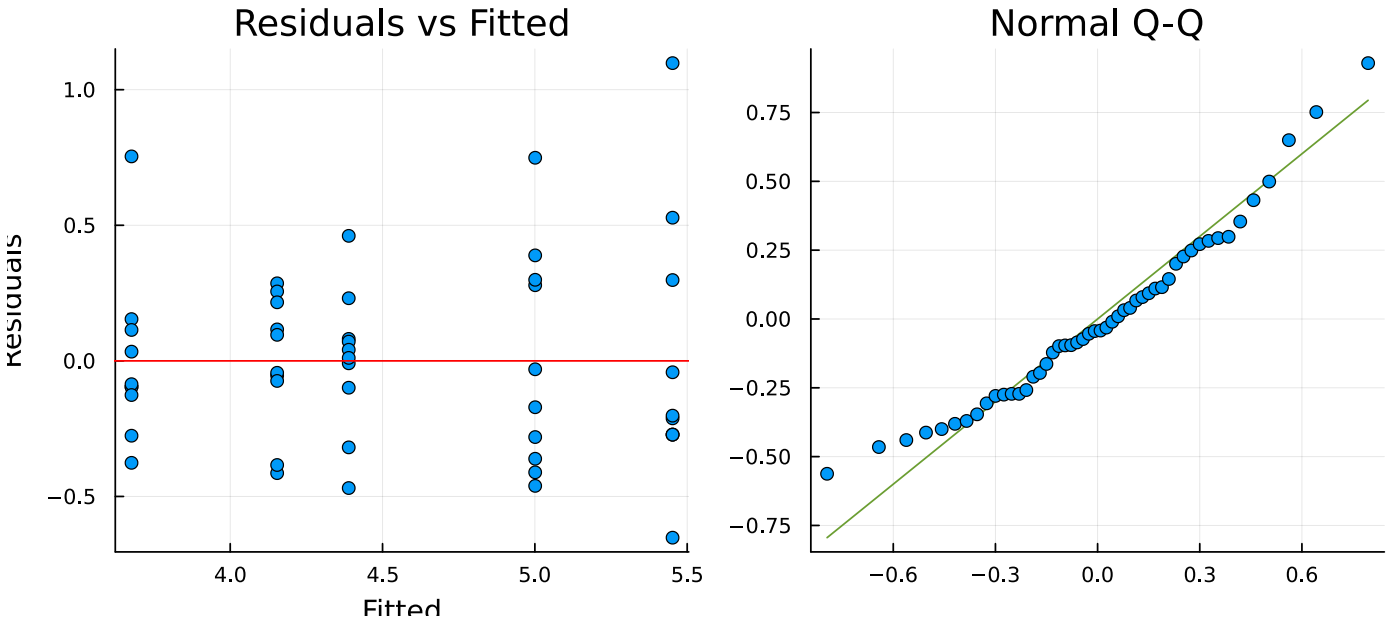
- Ayrshire: 4.5344 − 0.3804 = 4.154

- Canadian: 4.5344 − 0.1454 = 4.389
- Guernsey: 4.5344 + 0.4666 = 5.001
- Holstein-Fresian: 4.5344 − 0.8584 = 3.676
- Jersey: 4.5344 − (−0.3804 − 0.1454 + 0.4666 − 0.8584) = 4.5344 + 0.9176 = 5.452

The intercept (4.5344) is the average of the five breed means.

Part d)

Check the diagnostics for your model.




```
begin
    p1 = scatter(fitted(model), residuals(model), xlabel="Fitted", ylabel="Residuals",
    title="Residuals vs Fitted", legend=false)
    hline!(p1, [0], color=:red)
    p2 = qqnorm(residuals(model), title="Normal Q-Q", legend=false)
    plot(p1, p2, layout=(1, 2), size=(800, 350))
end
```

- Residuals vs fitted: points are spread around 0, with a bit more spread at higher fitted values (Guernsey, Jersey). No serious problem with constant variance.
- Q-Q plot: points follow the line fairly well, with a few large residuals at the top. Normality is reasonable.

The assumptions are reasonably met.

Part e)

Test for a difference between the breeds. (Make sure to run the full hypothesis test)

```
F-test: 2 models fitted on 50 observations
──────────────────────────────────────────────────────────────────────
     DOF   ΔDOF      SSR      ΔSSR       R²      ΔR²       F*   p(>F)
──────────────────────────────────────────────────────────────────────
[1]    2        25.4546            0.0000
[2]    6     4   5.8306  -19.6240  0.7709  0.7709  37.8639  <1e-13
──────────────────────────────────────────────────────────────────────
```

```
ftest(lm(@formula(Butterfat ~ 1), data1).model, model.model)
```

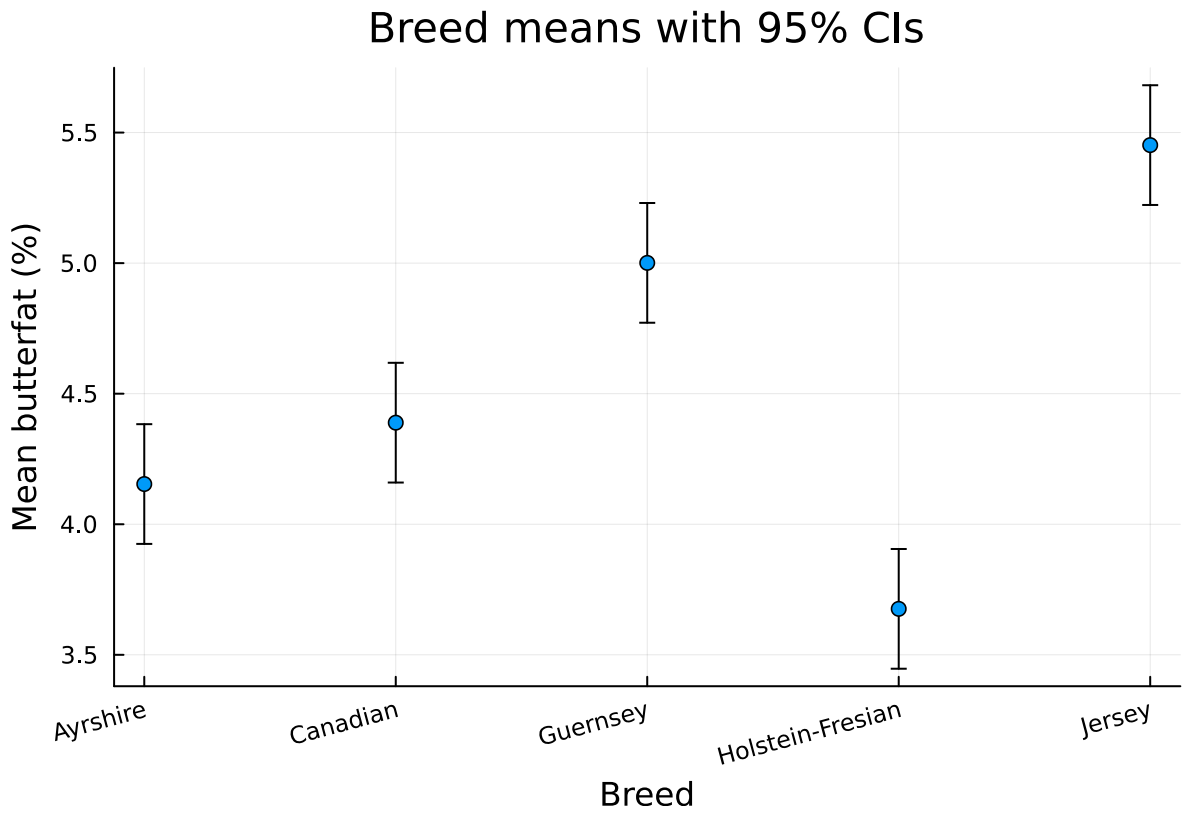
- H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (all breeds have the same mean butterfat)
- H_a: at least one breed mean is different.
- $\alpha = 0.05$
- Test statistic: $F = 37.86$ on 4 and 45 df
- p-value < 0.0001
- Decision: $p < \alpha$, so reject H_0.
- Conclusion: mean butterfat differs for at least one breed.

Part f)

Produce a plot to check for differences in butterfat between breeds. Which pairwise differences are not statistically signifcant? (You may also have to look at the summary)

	Breed	Mean	SD	Lower	Upper
1	"Ayrshire"	4.154	0.24627	3.92474	4.38326
2	"Canadian"	4.389	0.260532	4.15974	4.61826
3	"Guernsey"	5.001	0.408315	4.77174	5.23026
4	"Holstein-Fresian"	3.676	0.310133	3.44674	3.90526
5	"Jersey"	5.452	0.506377	5.22274	5.68126

```
begin
    # 95% CI for each breed mean: ȳ ± t* √(MSE/n)
    tcrit = quantile(TDist(dof_residual(model)), 0.975)
    se = sqrt(deviance(model) / dof_residual(model) / 10)
    ci = transform(stats, :Mean => (m -> m .- tcrit * se) => :Lower, :Mean => (m -> m
    .+ tcrit * se) => :Upper)
end
```




```
scatter(string.(ci.Breed), ci.Mean, yerror=tcrit * se, legend=false, xlabel="Breed",
ylabel="Mean butterfat (%)", title="Breed means with 95% CIs", xrotation=15)
```

Intervals that overlap are not significantly different: **Ayrshire and Canadian**, and **Guernsey and Jersey**. All other pairs do not overlap. Guernsey and Jersey only barely overlap, so that difference is borderline.